PADHAKU
Mathematics · Class 10 Mathematics



CHAPTER 5

Arithmetic Progressions

NCERT Exercise Solutions

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✨ Try each question yourself first, then check the answer. 💡 answers are samples: write your own.

Independent study aid by Beginers Point. Not an official NCERT publication.

In-text questions

5.2 **Explain why each of the lists (a) to (e) is an AP:**

(a) 147, 148, 149, ..., 157 (heights in cm)

(b) −3.1, −3.0, −2.9, −2.8, −2.7, −2.6, −2.5 (temperatures)

(c) 950, 900, 850, 800, ..., 50 (balance of loan)

(d) 200, 250, 300, 350, ..., 750 (cash prizes)

(e) 50, 100, 150, 200, ..., 500 (total savings)

✔ ANSWER

A list is an AP when every term is obtained by adding the same fixed number d to the previous term.

(a) 148 − 147 = 149 − 148 = 1. Same difference every time, so it is an AP with a = 147, **d = 1**.

(b) −3.0 − (−3.1) = −2.9 − (−3.0) = 0.1. So it is an AP with a = −3.1, **d = 0.1**.

(c) 900 − 950 = 850 − 900 = −50. So it is an AP with a = 950, **d = −50**.

(d) 250 − 200 = 300 − 250 = 50. So it is an AP with a = 200, **d = 50**.

(e) 100 − 50 = 150 − 100 = 50. So it is an AP with a = 50, **d = 50**.

All five are finite APs because each has a last term.

5.3 **Reena's salaries 8000, 8500, 9000, 9500, 10000, ... are in AP. Why?**

✔ ANSWER

8500 − 8000 = 9000 − 8500 = 9500 − 9000 = 10000 − 9500 = 500.

Each term is obtained by adding the fixed number 500 (the annual increment) to the previous term.

So the list is an **AP with a = 8000 and d = 500**.

📝 Exercise 5.1

1 (i) **Does the list make an AP, and why? The taxi fare after each km when the fare is ₹ 15 for the first km and ₹ 8 for each additional km.**

✔ ANSWER

Fare after 1 km, 2 km, 3 km, ... = 15, 15 + 8, 15 + 8 + 8, ... = 15, 23, 31, ...

23 − 15 = 31 − 23 = 8 (same fixed number every time).

Yes, it is an AP with a = 15 and d = 8.

1 (ii) **The amount of air present in a cylinder when a vacuum pump removes ¼ of the air remaining in the cylinder at a time.**

✔ ANSWER

Let the air at first be V. Each time ¼ is removed, so ¾ of the air remains.

Amounts: V, ¾V, $(¾)^2$V = 9V/16, $(¾)^3$V, ...

¾V − V = −¼V, but 9V/16 − ¾V = −3V/16. The differences are not the same.

No, it is not an AP (each term is multiplied by ¾, not added to by a fixed number).

1 (iii) **The cost of digging a well after every metre of digging, when it costs ₹ 150 for the first metre and rises by ₹ 50 for each subsequent metre.**

✔ ANSWER

Cost of 1st, 2nd, 3rd, ... metre = 150, 200, 250, ...

200 − 150 = 250 − 200 = 50.

Yes, it is an AP with a = 150 and d = 50.

1 (iv) **The amount of money in the account every year, when ₹ 10000 is deposited at compound interest at 8 % per annum.**

✔ ANSWER

Amount after 1, 2, 3, ... years = 10000(1.08), $10000(1.08)^2$, $10000(1.08)^3$, ... = 10800, 11664, 12597.12, ...

11664 − 10800 = 864, but 12597.12 − 11664 = 933.12. The differences are not equal.

No, it is not an AP (compound interest multiplies the amount by 1.08 each year).

2 **Write the first four terms of the AP when a and d are:**

(i) a = 10, d = 10

(ii) a = −2, d = 0

(iii) a = 4, d = −3

(iv) a = −1, d = ½

(v) a = −1.25, d = −0.25

✔ **ANSWER**

The AP is a, a + d, a + 2d, a + 3d.

(i) **10, 20, 30, 40**

(ii) **−2, −2, −2, −2**

(iii) 4, 4 − 3, 1 − 3, −2 − 3 = **4, 1, −2, −5**

(iv) −1, −1 + ½, −½ + ½, 0 + ½ = **−1, −½, 0, ½**

(v) **−1.25, −1.50, −1.75, −2.00**

3 **For the following APs, write the first term and the common difference:**

(i) 3, 1, −1, −3, ...

(ii) −5, −1, 3, 7, ...

(iii) 1/3, 5/3, 9/3, 13/3, ...

(iv) 0.6, 1.7, 2.8, 3.9, ...

✔ **ANSWER**

d = (second term) − (first term).

(i) a = 3, d = 1 − 3 = −2 → **a = 3, d = −2**

(ii) a = −5, d = −1 − (−5) = 4 → **a = −5, d = 4**

(iii) a = 1/3, d = 5/3 − 1/3 = 4/3 → **a = 1/3, d = 4/3**

(iv) a = 0.6, d = 1.7 − 0.6 = 1.1 → **a = 0.6, d = 1.1**

4 (i) **Is 2, 4, 8, 16, ... an AP? If yes, find d and write three more terms.**

✔ ANSWER

4 − 2 = 2, 8 − 4 = 4. Differences are not equal.

Not an AP.

4 (ii) **2, 5/2, 3, 7/2, ...**

✔ ANSWER

5/2 − 2 = ½, 3 − 5/2 = ½, 7/2 − 3 = ½.

Yes, an AP with d = ½.

Next three terms: 7/2 + ½ = 4, 4 + ½ = 9/2, 9/2 + ½ = 5 → **4, 9/2, 5**

4 (iii) **−1.2, −3.2, −5.2, −7.2, ...**

✔ ANSWER

−3.2 − (−1.2) = −2, −5.2 − (−3.2) = −2, −7.2 − (−5.2) = −2.

Yes, an AP with d = −2.

Next three terms: **−9.2, −11.2, −13.2**

4 (iv) **−10, −6, −2, 2, ...**

✔ ANSWER

−6 − (−10) = 4, −2 − (−6) = 4, 2 − (−2) = 4.

Yes, an AP with d = 4.

Next three terms: **6, 10, 14**

4 (v) **3, 3 + √2, 3 + 2√2, 3 + 3√2, ...**

✔ ANSWER

(3 + √2) − 3 = √2, (3 + 2√2) − (3 + √2) = √2, (3 + 3√2) − (3 + 2√2) = √2.

Yes, an AP with d = √2.

Next three terms: **3 + 4√2, 3 + 5√2, 3 + 6√2**

4 (vi) **0.2, 0.22, 0.222, 0.2222, ...**

✔ ANSWER

0.22 − 0.2 = 0.02, 0.222 − 0.22 = 0.002.

Differences are not equal.

Not an AP.

4 (vii) **0, −4, −8, −12, ...**

✔ ANSWER

−4 − 0 = −4, −8 − (−4) = −4, −12 − (−8) = −4.

Yes, an AP with d = −4.

Next three terms: **−16, −20, −24**

4 (viii) **$-\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}, \ldots$**

✔ ANSWER

$-\frac{1}{2} - (-\frac{1}{2}) = 0$ every time.

Yes, an AP with d = 0.

Next three terms: **$-\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}$**

4 (ix) **1, 3, 9, 27, ...**

✔ ANSWER

$3 - 1 = 2$, $9 - 3 = 6$. Differences are not equal.

Not an AP.

4 (x) **a, 2a, 3a, 4a, ...**

✔ ANSWER

$2a - a = a$, $3a - 2a = a$, $4a - 3a = a$.

Yes, an AP with d = a.

Next three terms: **5a, 6a, 7a**

4 (xi) **$a, a^2, a^3, a^4, \ldots$**

✔ ANSWER

$a^2 - a = a(a - 1)$, $a^3 - a^2 = a^2(a - 1)$. These are not equal (unless $a = 0$ or 1).

Not an AP.

4 (xii) **√2, √8, √18, √32, ...**

✔ ANSWER

√8 = 2√2, √18 = 3√2, √32 = 4√2. So the list is √2, 2√2, 3√2, 4√2, ...

2√2 − √2 = √2, 3√2 − 2√2 = √2, 4√2 − 3√2 = √2.

Yes, an AP with d = √2.

Next three terms: 5√2, 6√2, 7√2 = **√50, √72, √98**

4 (xiii) **√3, √6, √9, √12, ...**

✔ ANSWER

√6 − √3 = √3(√2 − 1) ≈ 0.73, √9 − √6 = 3 − √6 ≈ 0.55. Differences are not equal.

Not an AP.

4 (xiv) **$1^2, 3^2, 5^2, 7^2, ...$**

✔ ANSWER

The list is 1, 9, 25, 49, ...

9 − 1 = 8, 25 − 9 = 16. Differences are not equal.

Not an AP.

4 (xv) **$1^2, 5^2, 7^2, 73, ...$**

✔ ANSWER

The list is 1, 25, 49, 73, ...

25 − 1 = 24, 49 − 25 = 24, 73 − 49 = 24.

Yes, an AP with d = 24.

Next three terms: **97, 121, 145**

📝 Exercise 5.2

1 **Fill in the blanks (a = first term, d = common difference, a_n = nth term):**

	a	d	n	a_n
(i)	7	3	8	…
(ii)	−18	…	10	0
(iii)	…	−3	18	−5
(iv)	−18.9	2.5	…	3.6
(v)	3.5	0	105	…

✔ **ANSWER**

Use $a_n = a + (n − 1)d$.

(i) $a_8 = 7 + (8 − 1) \times 3 = 7 + 21 =$ **28**

(ii) $0 = −18 + (10 − 1)d \rightarrow 9d = 18 \rightarrow$ **d = 2**

(iii) $−5 = a + (18 − 1)(−3) = a − 51 \rightarrow$ **a = 46**

(iv) $3.6 = −18.9 + (n − 1) \times 2.5 \rightarrow (n − 1) \times 2.5 = 22.5 \rightarrow n − 1 = 9 \rightarrow$ **n = 10**

(v) $a_{105} = 3.5 + (105 − 1) \times 0 =$ **3.5**

2 (i) **30th term of the AP 10, 7, 4, … is (A) 97 (B) 77 (C) −77 (D) −87**

✔ **ANSWER**

(C) −77

$a = 10$, $d = 7 - 10 = -3$.

$a_{30} = 10 + (30 - 1)(-3) = 10 - 87 = -77$.

2 (ii) **11th term of the AP −3, −½, 2, ... is (A) 28 (B) 22 (C) −38 (D) −48½**

✔ ANSWER

(B) 22

$a = -3$, $d = -½ - (-3) = 5/2$.

$a_{11} = -3 + (11 - 1) \times 5/2 = -3 + 25 = 22$.

3 (i) **Find the missing terms: 2, ☐, 26**

✔ ANSWER

$a = 2$, $a_3 = 26 \rightarrow 2 + 2d = 26 \rightarrow d = 12$.

Missing term = 2 + 12 = **14**

3 (ii) **☐, 13, ☐, 3**

✔ ANSWER

$a_2 = a + d = 13$, $a_4 = a + 3d = 3$.

Subtracting: $2d = -10 \rightarrow d = -5$, so $a = 13 + 5 = 18$.

$a_3 = 13 - 5 = 8$.

Missing terms: **18, 8** (AP: 18, 13, 8, 3)

3 (iii) **5, ☐, ☐, 9½**

✔ ANSWER

$a = 5$, $a_4 = 5 + 3d = 9½ \rightarrow 3d = 4½ \rightarrow d = 1½$.

$a_2 = 5 + 1½ = 6½$, $a_3 = 6½ + 1½ = 8$.

Missing terms: **6½, 8**

3 (iv) **−4, □, □, □, □, 6**

✔ **ANSWER**

$a = -4$, $a_6 = -4 + 5d = 6 \rightarrow 5d = 10 \rightarrow d = 2$.

Missing terms: **−2, 0, 2, 4**

3 (v) **□, 38, □, □, □, −22**

✔ **ANSWER**

$a_2 = a + d = 38$, $a_6 = a + 5d = -22$.

Subtracting: $4d = -60 \rightarrow d = -15$, so $a = 38 + 15 = 53$.

$a_3 = 38 - 15 = 23$, $a_4 = 8$, $a_5 = -7$.

Missing terms: **53, 23, 8, −7**

4 **Which term of the AP 3, 8, 13, 18, … is 78?**

✔ **ANSWER**

$a = 3$, $d = 5$, $a_n = 78$.

$78 = 3 + (n - 1) \times 5 \rightarrow (n - 1) \times 5 = 75 \rightarrow n - 1 = 15 \rightarrow n = 16$.

78 is the 16th term.

5 (i) **Find the number of terms in the AP 7, 13, 19, ..., 205.**

✔ ANSWER

$a = 7$, $d = 6$, $a_n = 205$.

$205 = 7 + (n - 1) \times 6 \rightarrow (n - 1) \times 6 = 198 \rightarrow n - 1 = 33 \rightarrow$ **$n = 34$**

5 (ii) **Find the number of terms in the AP 18, 15½, 13, ..., −47.**

✔ ANSWER

$a = 18$, $d = 15½ - 18 = -5/2$, $a_n = -47$.

$-47 = 18 + (n - 1)(-5/2) \rightarrow (n - 1)(-5/2) = -65 \rightarrow n - 1 = 26 \rightarrow$ **$n = 27$**

6 **Check whether −150 is a term of the AP 11, 8, 5, 2, ...**

✔ ANSWER

$a = 11$, $d = -3$. Let $a_n = -150$.

$-150 = 11 + (n - 1)(-3) \rightarrow (n - 1)(-3) = -161 \rightarrow n - 1 = 161/3$.

$n = 164/3$, which is not a positive integer.

No, −150 is not a term of this AP.

7 **Find the 31st term of an AP whose 11th term is 38 and 16th term is 73.**

✔ ANSWER

$a_{11} = a + 10d = 38$ ... (1)

$a_{16} = a + 15d = 73$ ... (2)

(2) − (1): $5d = 35 \rightarrow d = 7$; $a = 38 - 70 = -32$.

$a_{31} = -32 + 30 \times 7 = -32 + 210 =$ **178**

8 **An AP consists of 50 terms of which the 3rd term is 12 and the last term is 106. Find the 29th term.**

✔ ANSWER

$a_3 = a + 2d = 12$ ... (1)

$a_{50} = a + 49d = 106$ ... (2)

(2) − (1): $47d = 94 \rightarrow d = 2$; $a = 12 - 4 = 8$.

$a_{29} = 8 + 28 \times 2 =$ **64**

9 **If the 3rd and 9th terms of an AP are 4 and −8 respectively, which term of this AP is zero?**

✔ ANSWER

$a + 2d = 4$ ... (1), $a + 8d = -8$ ... (2)

(2) − (1): $6d = -12 \rightarrow d = -2$; $a = 4 + 4 = 8$.

$a_n = 0 \rightarrow 8 + (n - 1)(-2) = 0 \rightarrow n - 1 = 4 \rightarrow n = 5$.

The 5th term is zero.

10 **The 17th term of an AP exceeds its 10th term by 7. Find the common difference.**

✔ ANSWER

$a_{17} - a_{10} = 7$

(a + 16d) − (a + 9d) = 7 → 7d = 7 → **d = 1**

11 **Which term of the AP 3, 15, 27, 39, … will be 132 more than its 54th term?**

✔ ANSWER

a = 3, d = 12.

a_{54} = 3 + 53 × 12 = 3 + 636 = 639.

Required term = 639 + 132 = 771.

771 = 3 + (n − 1) × 12 → (n − 1) × 12 = 768 → n − 1 = 64 → n = 65.

The 65th term.

12 **Two APs have the same common difference. The difference between their 100th terms is 100. What is the difference between their 1000th terms?**

✔ ANSWER

Let the first terms be a and b, common difference d (same for both).

$a_{100} - b_{100}$ = (a + 99d) − (b + 99d) = a − b = 100.

$a_{1000} - b_{1000}$ = (a + 999d) − (b + 999d) = a − b = **100**

13 **How many three-digit numbers are divisible by 7?**

✔ ANSWER

Three-digit numbers divisible by 7: 105, 112, 119, ..., 994 (an AP with a = 105, d = 7).

994 = 105 + (n − 1) × 7 → (n − 1) × 7 = 889 → n − 1 = 127 → n = 128.

128 numbers.

14 **How many multiples of 4 lie between 10 and 250?**

✔ ANSWER

Multiples of 4 between 10 and 250: 12, 16, 20, ..., 248 (a = 12, d = 4).

248 = 12 + (n − 1) × 4 → (n − 1) × 4 = 236 → n − 1 = 59 → n = 60.

60 multiples.

15 **For what value of n are the nth terms of the APs 63, 65, 67, ... and 3, 10, 17, ... equal?**

✔ ANSWER

First AP: $a_n = 63 + (n - 1) \times 2$. Second AP: $a_n = 3 + (n - 1) \times 7$.

63 + 2(n − 1) = 3 + 7(n − 1) → 60 = 5(n − 1) → n − 1 = 12 → **n = 13**

Check: 63 + 24 = 87 and 3 + 84 = 87. ✓

16 **Determine the AP whose third term is 16 and the 7th term exceeds the 5th term by 12.**

✔ ANSWER

$a_3 = a + 2d = 16$ ... (1)

$a_7 - a_5 = (a + 6d) - (a + 4d) = 2d = 12 \rightarrow d = 6$.

From (1): $a = 16 - 12 = 4$.

The AP is 4, 10, 16, 22, ...

17 **Find the 20th term from the last term of the AP 3, 8, 13, ..., 253.**

✔ ANSWER

Write the AP in reverse order: 253, 248, 243, ...; here $a = 253$, $d = -5$.

20th term from the last = a_{20} of the reversed AP = $253 + 19 \times (-5) = 253 - 95 =$ **158**

18 **The sum of the 4th and 8th terms of an AP is 24 and the sum of the 6th and 10th terms is 44. Find the first three terms.**

✔ ANSWER

$a_4 + a_8 = (a + 3d) + (a + 7d) = 2a + 10d = 24 \rightarrow a + 5d = 12$ ... (1)

$a_6 + a_{10} = (a + 5d) + (a + 9d) = 2a + 14d = 44 \rightarrow a + 7d = 22$ ... (2)

(2) − (1): $2d = 10 \rightarrow d = 5$; $a = 12 - 25 = -13$.

First three terms: −13, −8, −3

19 **Subba Rao started work in 1995 at an annual salary of ₹ 5000 and received an increment of ₹ 200 each year. In which year did his income reach ₹ 7000?**

✔ ANSWER

Salaries: 5000, 5200, 5400, ... (a = 5000, d = 200).

7000 = 5000 + (n − 1) × 200 → (n − 1) × 200 = 2000 → n − 1 = 10 → n = 11.

In the 11th year of service, i.e., in 2005 (1995 + 10).

20 **Ramkali saved ₹ 5 in the first week of a year and then increased her weekly savings by ₹ 1.75. If in the nth week her weekly savings become ₹ 20.75, find n.**

✔ ANSWER

a = 5, d = 1.75, a_n = 20.75.

20.75 = 5 + (n − 1) × 1.75 → (n − 1) × 1.75 = 15.75 → n − 1 = 9 → **n = 10**

Exercise 5.3

1 (i) **Find the sum of the AP 2, 7, 12, ..., to 10 terms.**

✔ ANSWER

a = 2, d = 5, n = 10.

$S_n = n/2\ [2a + (n - 1)d]$

$S_{10} = 10/2\ [4 + 9 \times 5] = 5 \times 49 =$ **245**

1 (ii) **−37, −33, −29, ..., to 12 terms.**

✔ ANSWER

a = −37, d = 4, n = 12.

$S_{12} = 12/2\ [-74 + 11 \times 4] = 6 \times (-74 + 44) = 6 \times (-30) =$ **−180**

1 (iii) **0.6, 1.7, 2.8, ..., to 100 terms.**

✔ ANSWER

a = 0.6, d = 1.1, n = 100.

$S_{100} = 100/2\ [1.2 + 99 \times 1.1] = 50 \times (1.2 + 108.9) = 50 \times 110.1 =$ **5505**

1 (iv) **1/15, 1/12, 1/10, ..., to 11 terms.**

✔ ANSWER

a = 1/15, d = 1/12 − 1/15 = (5 − 4)/60 = 1/60, n = 11.

$S_{11} = 11/2\ [2/15 + 10 \times 1/60] = 11/2\ [2/15 + 1/6] = 11/2 \times (4 + 5)/30 = 11/2 \times 9/30 = 11/2 \times 3/10 =$ **33/20**

2 (i) **Find the sum: 7 + 10½ + 14 + ... + 84**

✔ ANSWER

a = 7, d = 3½, last term l = 84.

84 = 7 + (n − 1) × 3.5 → (n − 1) × 3.5 = 77 → n − 1 = 22 → n = 23.

S = n/2 (a + l) = 23/2 × (7 + 84) = 23 × 91/2 = 2093/2 = **1046½**

2 (ii) **34 + 32 + 30 + ... + 10**

✔ ANSWER

a = 34, d = −2, l = 10.

10 = 34 + (n − 1)(−2) → n − 1 = 12 → n = 13.

S = 13/2 × (34 + 10) = 13 × 22 = **286**

2 (iii) **−5 + (−8) + (−11) + ... + (−230)**

✔ ANSWER

a = −5, d = −3, l = −230.

−230 = −5 + (n − 1)(−3) → (n − 1)(−3) = −225 → n − 1 = 75 → n = 76.

S = 76/2 × (−5 − 230) = 38 × (−235) = **−8930**

3 (i) **In an AP, given a = 5, d = 3, a_n = 50, find n and S_n.**

✔ ANSWER

50 = 5 + (n − 1) × 3 → n − 1 = 15 → **n = 16**

S_n = n/2 (a + l) = 16/2 × (5 + 50) = 8 × 55 = **440**

3 (ii) **Given $a = 7$, $a_{13} = 35$, find d and S_{13}.**

✔ ANSWER

$35 = 7 + 12d \rightarrow 12d = 28 \rightarrow \mathbf{d = 7/3}$

$S_{13} = 13/2 \times (7 + 35) = 13 \times 21 = \mathbf{273}$

3 (iii) **Given $a_{12} = 37$, $d = 3$, find a and S_{12}.**

✔ ANSWER

$37 = a + 11 \times 3 \rightarrow \mathbf{a = 4}$

$S_{12} = 12/2 \times (4 + 37) = 6 \times 41 = \mathbf{246}$

3 (iv) **Given $a_3 = 15$, $S_{10} = 125$, find d and a_{10}.**

✔ ANSWER

$a + 2d = 15$ ... (1)

$S_{10} = 10/2\ [2a + 9d] = 125 \rightarrow 2a + 9d = 25$ ... (2)

$2 \times$ (1): $2a + 4d = 30$ ... (3). (2) − (3): $5d = -5 \rightarrow$ **$d = -1$**; $a = 15 + 2 = 17$.

$a_{10} = 17 + 9 \times (-1) = \mathbf{8}$

3 (v) **Given $d = 5$, $S_9 = 75$, find a and a_9.**

✔ ANSWER

$S_9 = 9/2\ [2a + 8 \times 5] = 75 \rightarrow 2a + 40 = 150/9 = 50/3$

$2a = 50/3 - 40 = (50 - 120)/3 = -70/3 \rightarrow$ **$a = -35/3$**

$a_9 = -35/3 + 8 \times 5 = -35/3 + 40 = (-35 + 120)/3 =$ **85/3**

3 (vi) **Given a = 2, d = 8, S_n = 90, find n and a_n.**

✔ ANSWER

$90 = n/2\ [4 + (n - 1) \times 8] = n/2\ (8n - 4) = n(4n - 2)$

$4n^2 - 2n - 90 = 0 \rightarrow 2n^2 - n - 45 = 0$

$(2n + 9)(n - 5) = 0 \rightarrow n = 5$ (n cannot be −9/2) → **n = 5**

$a_5 = 2 + 4 \times 8 =$ **34**

3 (vii) **Given a = 8, a_n = 62, S_n = 210, find n and d.**

✔ ANSWER

$S_n = n/2\ (a + a_n) \rightarrow 210 = n/2\ (8 + 62) = 35n \rightarrow$ **n = 6**

$a_6 = 62 \rightarrow 8 + 5d = 62 \rightarrow 5d = 54 \rightarrow$ **d = 54/5**

3 (viii) **Given a_n = 4, d = 2, S_n = −14, find n and a.**

✔ ANSWER

$a_n = a + (n - 1) \times 2 = 4 \rightarrow a = 4 - 2(n - 1) = 6 - 2n$ ... (1)

$S_n = n/2\ (a + 4) = -14 \rightarrow n(6 - 2n + 4) = -28 \rightarrow n(10 - 2n) = -28 \rightarrow n(5 - n) = -14$

$n^2 - 5n - 14 = 0 \rightarrow (n - 7)(n + 2) = 0 \rightarrow$ **n = 7**

From (1): a = 6 − 14 = **−8**

3 (ix) **Given a = 3, n = 8, S = 192, find d.**

✔ ANSWER

192 = 8/2 [6 + 7d] = 4(6 + 7d) → 6 + 7d = 48 → 7d = 42 → **d = 6**

3 (x) **Given l = 28, S = 144, and there are 9 terms in all. Find a.**

✔ ANSWER

S = n/2 (a + l) → 144 = 9/2 (a + 28) → a + 28 = 32 → **a = 4**

4 **How many terms of the AP 9, 17, 25, ... must be taken to give a sum of 636?**

✔ ANSWER

a = 9, d = 8, S_n = 636.

636 = n/2 [18 + (n − 1) × 8] = n/2 (8n + 10) = n(4n + 5)

$4n^2 + 5n - 636 = 0$

n = [−5 ± √(25 + 4 × 4 × 636)]/8 = [−5 ± √10201]/8 = (−5 ± 101)/8

n = 96/8 = 12 or n = −106/8 (rejected, n must be a positive integer).

12 terms

5 **The first term of an AP is 5, the last term is 45 and the sum is 400. Find the number of terms and the common difference.**

✔ ANSWER

$S = n/2\ (a + l) \rightarrow 400 = n/2\ (5 + 45) = 25n \rightarrow$ **n = 16**

$l = a + (n - 1)d \rightarrow 45 = 5 + 15d \rightarrow 15d = 40 \rightarrow$ **d = 8/3**

6 **The first and last terms of an AP are 17 and 350. If d = 9, how many terms are there and what is their sum?**

✔ ANSWER

$350 = 17 + (n - 1) \times 9 \rightarrow (n - 1) \times 9 = 333 \rightarrow n - 1 = 37 \rightarrow$ **n = 38**

$S_{38} = 38/2\ (17 + 350) = 19 \times 367 =$ **6973**

7 **Find the sum of the first 22 terms of an AP in which d = 7 and the 22nd term is 149.**

✔ ANSWER

$a_{22} = a + 21 \times 7 = 149 \rightarrow a = 149 - 147 = 2.$

$S_{22} = 22/2\ (2 + 149) = 11 \times 151 =$ **1661**

8 **Find the sum of the first 51 terms of an AP whose second and third terms are 14 and 18 respectively.**

✔ ANSWER

$d = a_3 - a_2 = 18 - 14 = 4$; $a = a_2 - d = 14 - 4 = 10$.

$S_{51} = 51/2\ [20 + 50 \times 4] = 51/2 \times 220 = 51 \times 110 =$ **5610**

9 **If the sum of the first 7 terms of an AP is 49 and that of 17 terms is 289, find the sum of the first n terms.**

✔ ANSWER

$S_7 = 7/2\ [2a + 6d] = 49 \rightarrow 2a + 6d = 14 \rightarrow a + 3d = 7$ ... (1)

$S_{17} = 17/2\ [2a + 16d] = 289 \rightarrow 2a + 16d = 34 \rightarrow a + 8d = 17$ ... (2)

(2) − (1): $5d = 10 \rightarrow d = 2$; $a = 7 - 6 = 1$.

$S_n = n/2\ [2 \times 1 + (n - 1) \times 2] = n/2 \times 2n =$ **n^2**

10 (i) **Show that $a_1, a_2, \ldots, a_n, \ldots$ form an AP where $a_n = 3 + 4n$. Also find the sum of the first 15 terms.**

✔ ANSWER

$a_1 = 7$, $a_2 = 11$, $a_3 = 15$, ...

$a_{n+1} - a_n = [3 + 4(n + 1)] - [3 + 4n] = 4$, the same for every n.

So the list is an **AP with a = 7, d = 4**.

$S_{15} = 15/2\ [14 + 14 \times 4] = 15/2 \times 70 = 15 \times 35 =$ **525**

10 (ii) **$a_n = 9 - 5n$. Show it is an AP and find S_{15}.**

✔ ANSWER

$a_1 = 4$, $a_2 = -1$, $a_3 = -6$, ...

$a_{n+1} - a_n = [9 - 5(n + 1)] - [9 - 5n] = -5$, the same for every n.

So the list is an **AP with a = 4, d = −5**.

$S_{15} = 15/2\ [8 + 14 \times (-5)] = 15/2 \times (-62) = 15 \times (-31) =$ **−465**

11 **If the sum of the first n terms of an AP is $4n - n^2$, what is the first term (S_1)? What is the sum of the first two terms? What is the second term? Similarly, find the 3rd, the 10th and the nth terms.**

✔ ANSWER

$S_n = 4n - n^2$

$S_1 = 4 - 1 = 3 \rightarrow$ **first term $a_1 = 3$**

$S_2 = 8 - 4 =$ **4**

$a_2 = S_2 - S_1 = 4 - 3 =$ **1**

$S_3 = 12 - 9 = 3 \rightarrow a_3 = S_3 - S_2 = 3 - 4 =$ **−1**

$S_{10} = 40 - 100 = -60$, $S_9 = 36 - 81 = -45 \rightarrow a_{10} = -60 - (-45) =$ **−15**

$a_n = S_n - S_{n-1} = (4n - n^2) - [4(n - 1) - (n - 1)^2] = 4n - n^2 - 4n + 4 + n^2 - 2n + 1 =$ **5 − 2n**

Check: $a_{10} = 5 - 20 = -15$. ✓

12 **Find the sum of the first 40 positive integers divisible by 6.**

✔ ANSWER

6, 12, 18, ...; a = 6, d = 6, n = 40.

$S_{40} = 40/2\ [12 + 39 \times 6] = 20 \times (12 + 234) = 20 \times 246 =$ **4920**

13 **Find the sum of the first 15 multiples of 8.**

✔ ANSWER

8, 16, 24, ..., 120; a = 8, l = 15 × 8 = 120, n = 15.

$S_{15} = 15/2\ (8 + 120) = 15 \times 64 =$ **960**

14 **Find the sum of the odd numbers between 0 and 50.**

✔ ANSWER

1, 3, 5, ..., 49; a = 1, d = 2, l = 49.

$49 = 1 + (n - 1) \times 2 \rightarrow n = 25$.

$S_{25} = 25/2\ (1 + 49) = 25 \times 25 =$ **625**

15 **A contract specifies a penalty for delay: ₹ 200 for the first day, ₹ 250 for the second day, ₹ 300 for the third day, etc., each day ₹ 50 more than the preceding day. How much penalty for a delay of 30 days?**

✔ ANSWER

Penalties: 200, 250, 300, ...; a = 200, d = 50, n = 30.

$S_{30} = 30/2\ [400 + 29 \times 50] = 15 \times (400 + 1450) = 15 \times 1850 = 27750$

Total penalty = ₹ 27,750

16 **A sum of ₹ 700 is to be used to give seven cash prizes. If each prize is ₹ 20 less than its preceding prize, find the value of each prize.**

✔ ANSWER

Let the first prize be ₹ a. The prizes form an AP with d = −20, n = 7, $S_7 = 700$.

$700 = 7/2\ [2a + 6 \times (-20)] = 7(a - 60) \rightarrow a - 60 = 100 \rightarrow a = 160$

Prizes: ₹ 160, ₹ 140, ₹ 120, ₹ 100, ₹ 80, ₹ 60, ₹ 40

17 **Each section of each class plants as many trees as its class number (Class I: 1 tree, Class II: 2 trees, ... Class XII: 12 trees). There are three sections of each class. How many trees will be planted?**

✔ ANSWER

Trees per class (one section): 1, 2, 3, ..., 12 → sum = 12/2 (1 + 12) = 78.

Three sections of each class: 3 × 78 = 234

234 trees

18 **A spiral is made of successive semicircles with centres alternately at A and B, of radii 0.5 cm, 1.0 cm, 1.5 cm, 2.0 cm, ... What is the total length of a spiral made of thirteen consecutive semicircles? (π = 22/7)**

✔ ANSWER

Length of a semicircle of radius r = πr.

Lengths: π(0.5), π(1.0), π(1.5), ... — an AP with a = 0.5π, d = 0.5π, n = 13.

Total = π × [0.5 + 1.0 + ... + 6.5] = π × 13/2 × (0.5 + 6.5) = π × 13/2 × 7 = 45.5π

= 22/7 × 45.5 = 22 × 6.5 = 143

Total length = 143 cm

19 **200 logs are stacked: 20 in the bottom row, 19 in the next, 18 in the next, and so on. In**

how many rows are the 200 logs placed and how many logs are in the top row?

✔ ANSWER

$a = 20$, $d = -1$, $S_n = 200$.

$200 = n/2\,[40 + (n - 1)(-1)] = n/2\,(41 - n) \rightarrow 400 = 41n - n^2$

$n^2 - 41n + 400 = 0 \rightarrow (n - 16)(n - 25) = 0 \rightarrow n = 16$ or 25

If $n = 25$: $a_{25} = 20 + 24 \times (-1) = -4$, which is not possible. So $n = 16$.

$a_{16} = 20 + 15 \times (-1) = 5$

16 rows; 5 logs in the top row

20 **In a potato race, a bucket is 5 m from the first potato and the other potatoes are 3 m apart in a line; there are 10 potatoes. The competitor runs from the bucket to each potato and back, one at a time. What is the total distance run?**

✔ ANSWER

Distances to the potatoes: 5, 8, 11, ... m. Each potato is brought back, so the run for each is doubled.

Distances run: 2×5, 2×8, 2×11, ... = 10, 16, 22, ...; $a = 10$, $d = 6$, $n = 10$.

$S_{10} = 10/2\,[20 + 9 \times 6] = 5 \times 74 = 370$

Total distance = 370 m

📝 Exercise 5.4 (Optional)

1 **Which term of the AP 121, 117, 113, ... is its first negative term?**

✔ ANSWER

$a = 121$, $d = -4$. $a_n = 121 + (n - 1)(-4) = 125 - 4n$.

For $a_n < 0$: $125 - 4n < 0 \rightarrow n > 125/4 = 31.25$

The smallest such integer is $n = 32$. ($a_{32} = 125 - 128 = -3$; $a_{31} = 1$.)

The 32nd term is the first negative term.

2 **The sum of the third and seventh terms of an AP is 6 and their product is 8. Find the sum of the first sixteen terms.**

✔ ANSWER

$a_3 + a_7 = (a + 2d) + (a + 6d) = 2a + 8d = 6 \rightarrow a + 4d = 3 \rightarrow a = 3 - 4d$

$a_3 \times a_7 = (a + 2d)(a + 6d) = 8 \rightarrow (3 - 2d)(3 + 2d) = 8 \rightarrow 9 - 4d^2 = 8 \rightarrow d^2 = ¼ \rightarrow d = \pm ½$

Case $d = ½$: $a = 3 - 2 = 1$; $S_{16} = 16/2\,[2 + 15 \times ½] = 8 \times 9.5 =$ **76**

Case $d = -½$: $a = 3 + 2 = 5$; $S_{16} = 16/2\,[10 + 15 \times (-½)] = 8 \times 2.5 =$ **20**

S_{16} = 76 or 20

3 **A ladder has rungs 25 cm apart. The rungs decrease uniformly in length from 45 cm at**

the bottom to 25 cm at the top. If the top and bottom rungs are 2½ m apart, what is the length of wood required for the rungs?

✔ ANSWER

Distance between top and bottom rungs = 2½ m = 250 cm; rungs are 25 cm apart.

Number of rungs = 250/25 + 1 = 11.

Lengths form an AP with a = 45, l = 25, n = 11.

Total length = 11/2 (45 + 25) = 11 × 35 = 385

385 cm of wood

4 **Houses of a row are numbered 1 to 49. Show that there is a value of x such that the sum of the numbers of the houses before house x equals the sum of the numbers of the houses after it. Find x.**

✔ ANSWER

Sum of 1 to n is $S_n = n(n + 1)/2$.

Sum before house x = $S_{x-1} = (x - 1)x/2$

Sum after house x = $S_{49} - S_x = 49 \times 50/2 - x(x + 1)/2 = 1225 - x(x + 1)/2$

Condition: $(x - 1)x/2 = 1225 - x(x + 1)/2 \rightarrow x^2 - x = 2450 - x^2 - x \rightarrow 2x^2 = 2450 \rightarrow x^2 = 1225$

x = 35 (positive value). So such a value exists: **x = 35**

Check: sum of 1 to 34 = 595; sum of 36 to 49 = 1225 − 630 = 595. ✓

5 **A terrace at a football ground has 15 steps, each 50 m long, built of solid concrete. Each step has a rise of ¼ m and a tread of ½ m. Calculate the total volume of concrete required.**

✔ ANSWER

Each step is a cuboid 50 m long and ½ m wide (tread). The 1st step is ¼ m high, the 2nd is 2 × ¼ = ½ m high, the 3rd is ¾ m high, and so on (each step is built from the ground).

Volume of 1st step = ¼ × ½ × 50 = 25/4 m^3

Volume of 2nd step = ½ × ½ × 50 = 50/4 m^3, 3rd = 75/4 m^3, ... — an AP with a = 25/4, d = 25/4, n = 15.

Total = 15/2 [2 × 25/4 + 14 × 25/4] = 15/2 × 16 × 25/4 = 15 × 50 = 750

Total volume = 750 m^3

🎉 **Chapter 5 done! Revise the answers once more tomorrow.**